Seasonal adjustment of economic data
Outliers and modelling the Covid-19 pandemic

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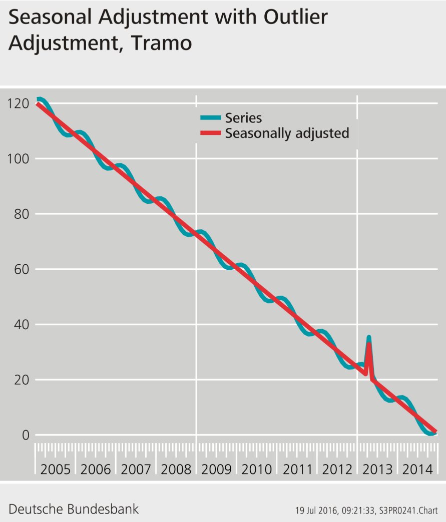
Seasonal adjustment, 06 November 2025

What are outliers?
Intuition

Outliers are

- Persistent changes
  - Changes in methodology
  - Economic policy
- Temporary changes
  - Atypical events
  - Strikes

Why do we need outlier adjustment? (I/VI)
TramoSeats, with outlier adjustment

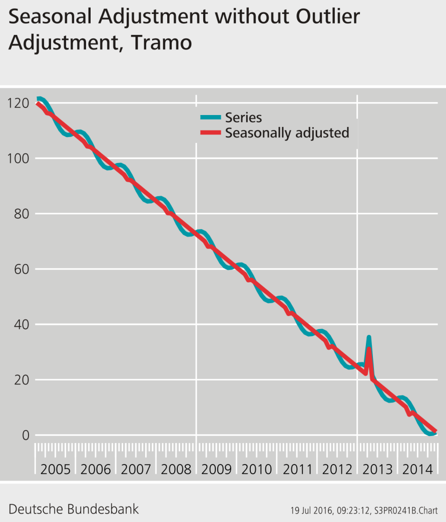


ARIMA Model:
(0 1 0)(0 1 0)

	Coef	T-Stat
AO(4-2013)	11.8	> 100

Why do we need outlier adjustment? (II/VI)

TramoSeats, without outlier adjustment



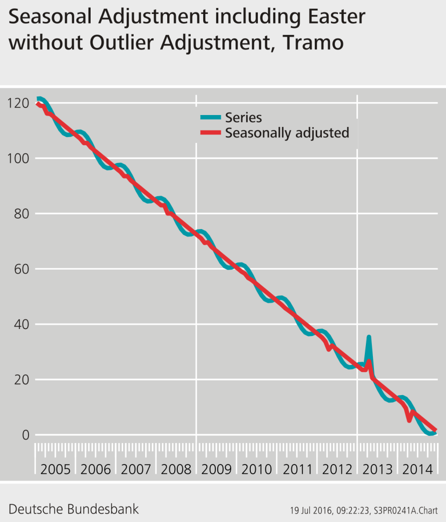
ARIMA Model

(0 1 1)(0 1 1)

	Coef	T-Stat
θ	-1.0	< -100
Θ	-0.8	-15.4

Why do we need outlier adjustment? (III/VI)

TramoSeats, without outlier adjustment, including Easter

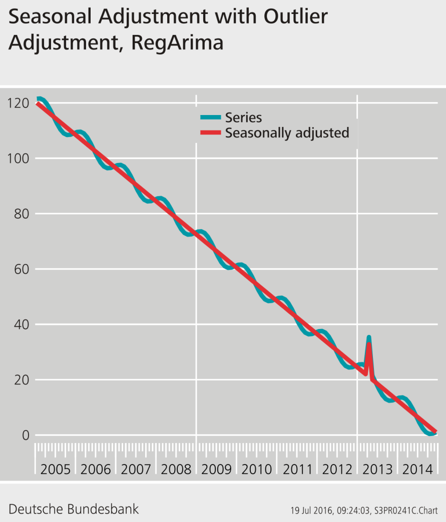


<u>ARIMA Model</u>

(0 1 1)(1 1 0)

	Coef	T-Stat
Easter	-1.9	-4.4
θ	-1.0	-34.2
Φ	0.8	7.7

Why do we need outlier adjustment? (IV/VI)
X13, with outlier adjustment

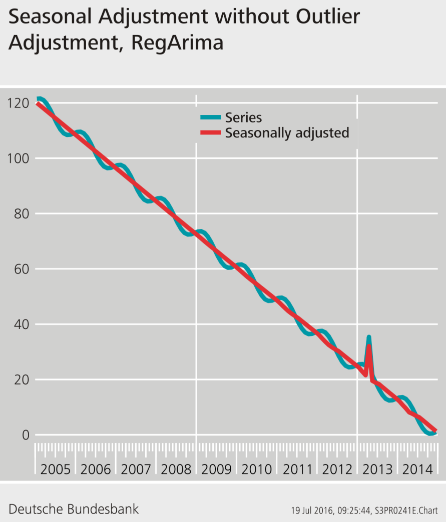


ARIMA Model
(1 1 0)(0 1 0)

	Coeff	T-Stat
ϕ	0.5	6.1
AO(4-2013)	11.8	> 100

Why do we need outlier adjustment? (V/VI)

X13, without outlier adjustment



ARIMA Model

(0 0 0)(0 1 1) + mean

	Coef	T-Stat
μ	-12.0	< -100
Θ	-0.9	-10.4

Why do we need outlier adjustment? (VI/VI)

Undetected outliers can have an influence on

- Parameter estimates
  - $\rightsquigarrow$ Forecast
  - $\rightsquigarrow$ Calendar effects

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- Parameter estimates
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  - $\rightsquigarrow$ Calendar effects
- Seasonal factors*

* *Mitigated by extreme value detection in X-11*

Why do we need outlier adjustment? (VI/VI)

Undetected outliers can have an influence on

- Parameter estimates
  - $\rightsquigarrow$ Forecast
  - $\rightsquigarrow$ Calendar effects
- Seasonal factors*
- Interpretation of results

* *Mitigated by extreme value detection in X-11*

The model (I/II)
Model without outliers

The usual ARIMA Model without outliers is given by

$$\phi_p(B)\,\Phi_P(B^\tau)\nabla^d\,\nabla_\tau^D y_t = \theta_q(B)\,\Theta_Q(B^\tau)\,\varepsilon_t \tag{1}$$

The model (II/II)
Model with outliers

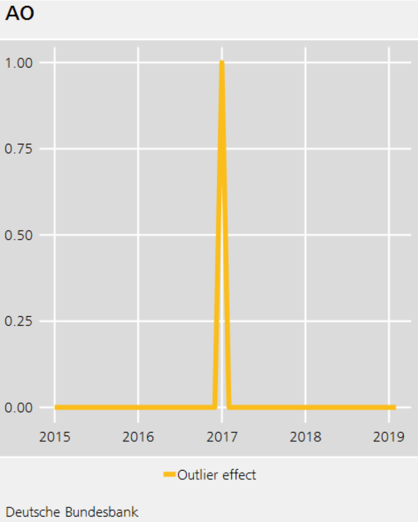
The ARIMA model with multiple outliers is obtained by

$$z_t = y_t + \sum_{j=1}^{J} \omega_j \nu_j \xi_t^{(t_j^\star)} \tag{2}$$

with

$$\begin{aligned} \omega_j &= \text{Outlier effect} \\ \nu_j &= \text{Dynamic behaviour of outlier} \\ \xi_t^{(t_j^\star)} &= \text{Indicator for occurence of outlier at } t_j^\star \end{aligned}$$

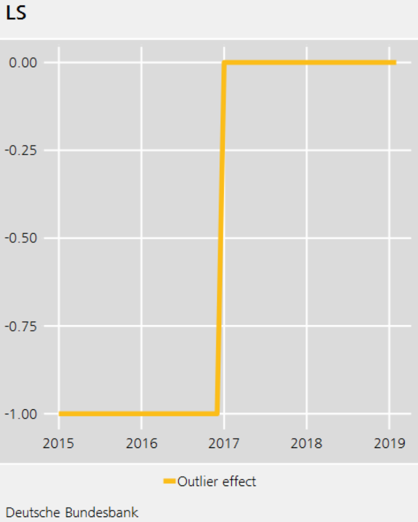
Types of outliers (I/IV)
Additive outlier (AO)



$$\nu_{AO} = 1 \tag{3}$$

$$z_t = y_t + \omega_{AO}\xi_t^{(t^\star)}$$

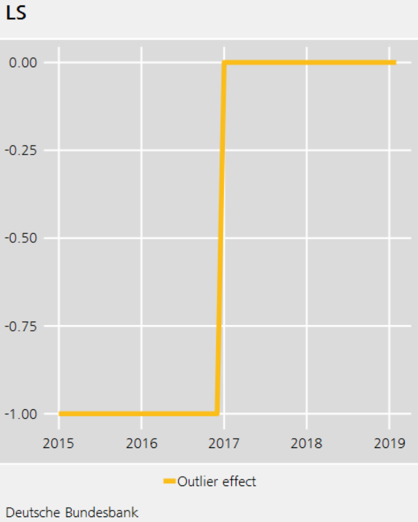
Types of outliers (II/IV)
Level shift (LS)



$$\nu_{LS} = \frac{1}{1-B} \tag{4}$$

$$z_t = y_t + \omega_{LS} \frac{1}{1-B} \xi_t^{(t^\star)}$$

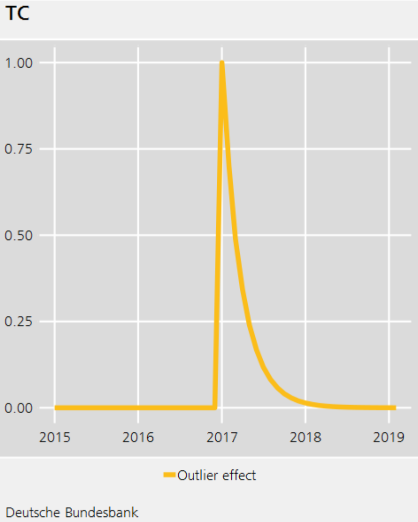
Types of outliers (II/IV)
Level shift (LS)



$$\nu_{LS} = \frac{1}{1-B} \tag{4}$$

$$\begin{aligned} z_t &= y_t + \omega_{LS} \frac{1}{1-B} \xi_t^{(t^\star)} \\ &= y_t + \omega_{LS} (1 + B + B^2 \ldots) \xi_t^{(t^\star)} \end{aligned}$$

Types of outliers (III/IV)
Temporary change (TC)



$$\nu_{TC} = \frac{1}{1-\delta B} \tag{5}$$

$$\begin{aligned} z_t &= y_t + \omega_{TC}\frac{1}{1-\delta B}\xi_t^{(t^\star)} \\ &= y_t + \omega_{TC}\sum_{i=0}^{\infty}\delta^i \xi_{t-i}^{(t^\star)} \end{aligned}$$

Exercise

Open JDemetra+ and run a seasonal adjustment of a time series of your choice.

- Which outliers were found automatically?
- Set an AO or an LS at the beginning of the Covid-19 pandemic. Is it significant?

Digression: Decay rate in TC

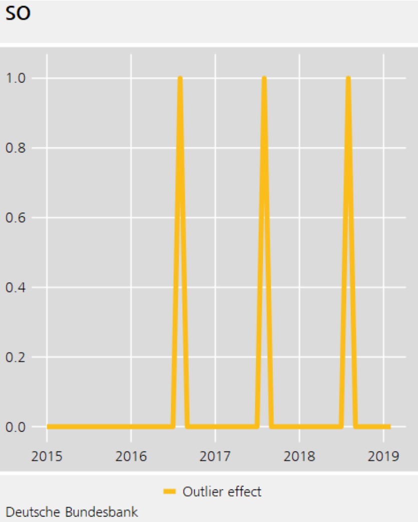
Default in JDemetra+

- $\delta = 0.7$

During Covid-19 pandemic

- Some agencies in the world experimented with different values of δ

Types of outliers (IV/IV)
Seasonal outlier (SO)



$$\nu_{SLS} = \frac{1}{1 - B^S} \tag{6}$$

$$z_t = y_t + \omega_{SLS} \frac{1}{1 - B^S} \xi_t^{(t^\star)}$$

Specification

Automatic detection

- Forward-backward-algorithm

User choice

- Data $\rightsquigarrow$ Expert knowledge, graphical analysis

Remarks

- Current end $\rightsquigarrow$ Turning point vs. (additive) outlier
- Significance $\rightsquigarrow$ Economic vs. statistical

The automatic algorithm
Gomez and Maravall (2001), Chen and Liu (1993)

Stage I: Outlier detection (assuming independent outliers)

The automatic algorithm
Gomez and Maravall (2001), Chen and Liu (1993)

Stage I: Outlier detection (assuming independent outliers)

Stage II: Joint outlier estimation and elimination

The automatic algorithm
Gomez and Maravall (2001), Chen and Liu (1993)

Stage I: Outlier detection (assuming independent outliers)

Stage II: Joint outlier estimation and elimination

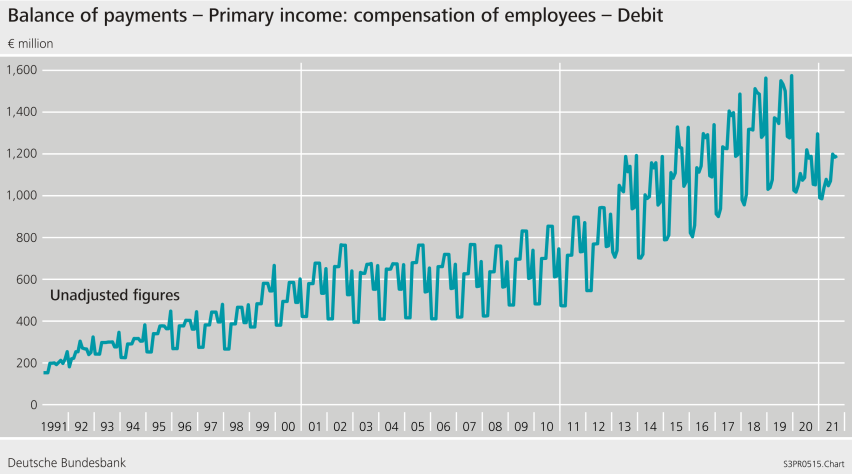
Repeat until convergence

Exercise

Using the time series you used before:

- Change the critical value to 3.5. Are more outliers found?
- Change the specification so that only Level Shifts are found. What happens?

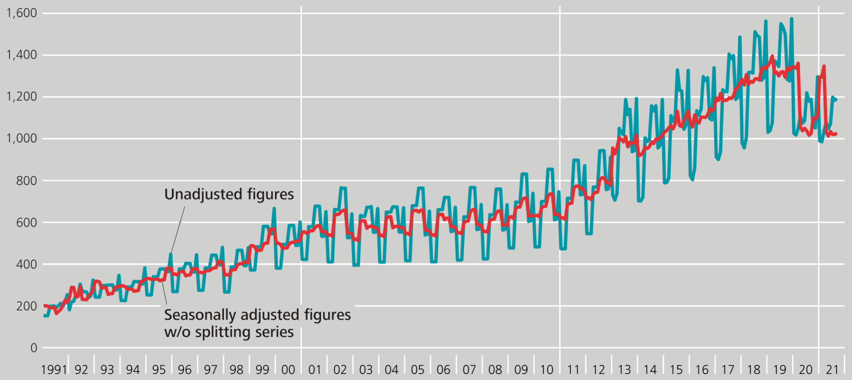
Seasonal breaks



Seasonal breaks

Balance of payments – Primary income: compensation of employees – Debit

€ million



Deutsche Bundesbank S3PR0515A.Chart

Seasonal breaks

Characteristics

- Abrupt changes in the seasonal pattern

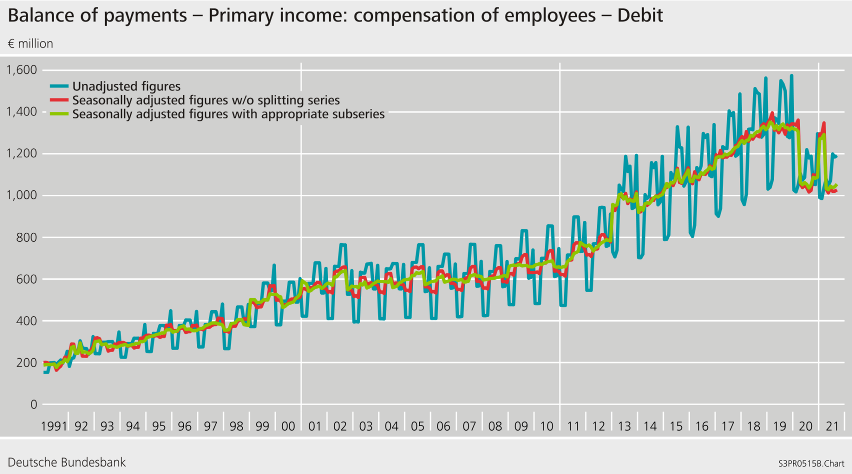
Challenges

- Decision between single event and new pattern
- Seasonal filters are better suited to model gradual changes

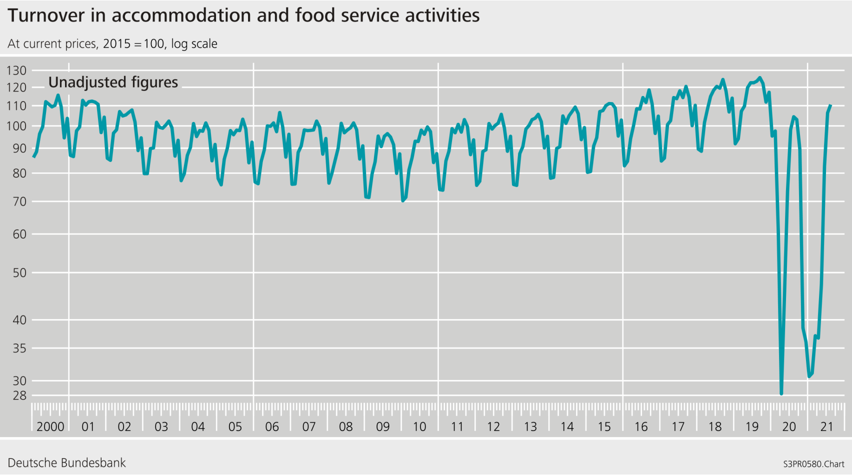
Potential remedies

- Split series for seasonal adjustment
- Use short seasonal filters
- Model as seasonal outliers

Seasonal breaks

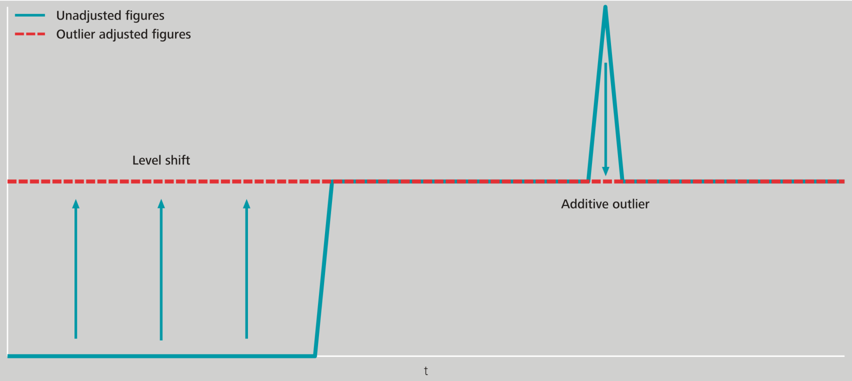


Covid-19 pandemic



Correction

Basic principle of temporary outlier removal



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S3PR0092.Chart

Estimated effects

Turnover in accommodation and food service activities At current prices, 2015 = 100			
Type	Date	Parameter estimate	t-value
Level shift	Jan 2009	-0.0573	-5.2
	Mar 2020	-0.5883	-39.8
	Apr 2020	-0.7994	-47.1
	May 2020	0.3606	21.3
	Jun 2020	0.4943	29.1
	Jul 2020	0.2660	15.7
	Aug 2020	0.0484	3.2
	Oct 2020	-0.1348	-8.8
	Nov 2020	-0.7162	-42.2
	Dec 2020	-0.1240	-7.3
	Jan 2021	0.0366	2.4
	Mar 2021	0.0551	3.5
	Apr 2021	-0.0603	-3.4
	May 2021	0.1598	9.1
	Jun 2021	0.5657	32.1
	Jul 2021	0.2100	11.9
	Aug 2021	0.0548	3.1

Outlier component

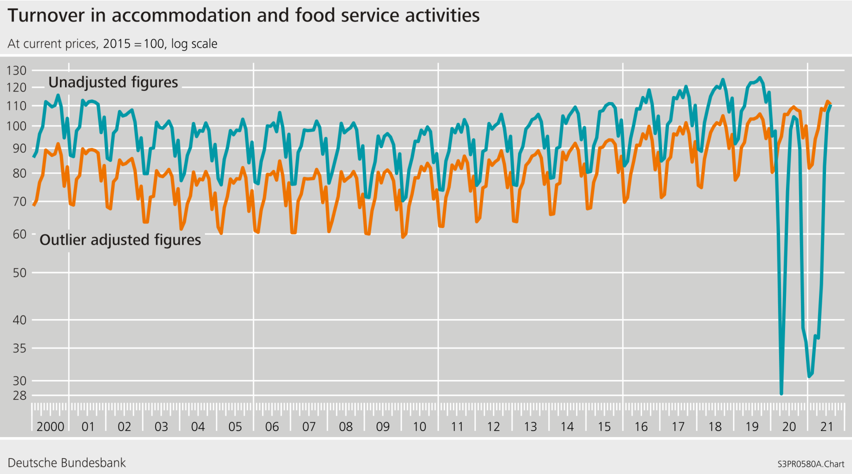
Turnover in accommodation and food service activities

At current prices, 2015 = 100, regARIMA combined outlier component [1)]

	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
2000	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26
2001	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26
...												
2008	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26
2009	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19
2010	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19
...												
2019	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19	1.19
2020	1.19	1.19	0.66	0.30	0.43	0.70	0.91	0.96	0.96	0.83	0.41	0.36
2021	0.37	0.37	0.40	0.37	0.47	0.77	0.95	1.00				

1 Blue scaling factors correspond to dates of level shifts.

Temporary removal



Treatment of Covid-19 pandemic effects (I/III)

Data issue: treatment of end point of the series

- Modelling based on statistical criteria and economic information
- AO and LS cannot be distinguished at last data point
- Remedy: use projected seasonal factors

Treatment of Covid-19 pandemic effects (II/III)

Pay attention to decomposition scheme

- Additive: near zero values may cause negative adjusted values
- Multiplicative: does proportionality assumption still hold?

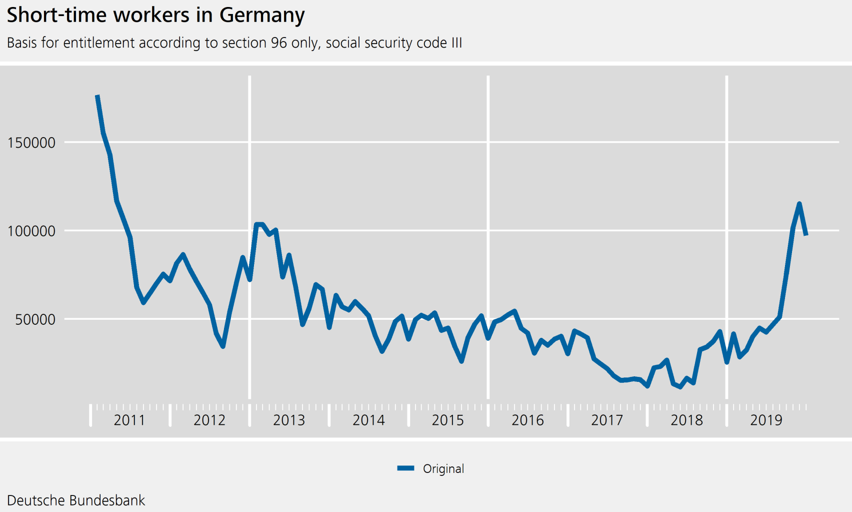
Treatment of Covid-19 pandemic effects (III/III)

More information

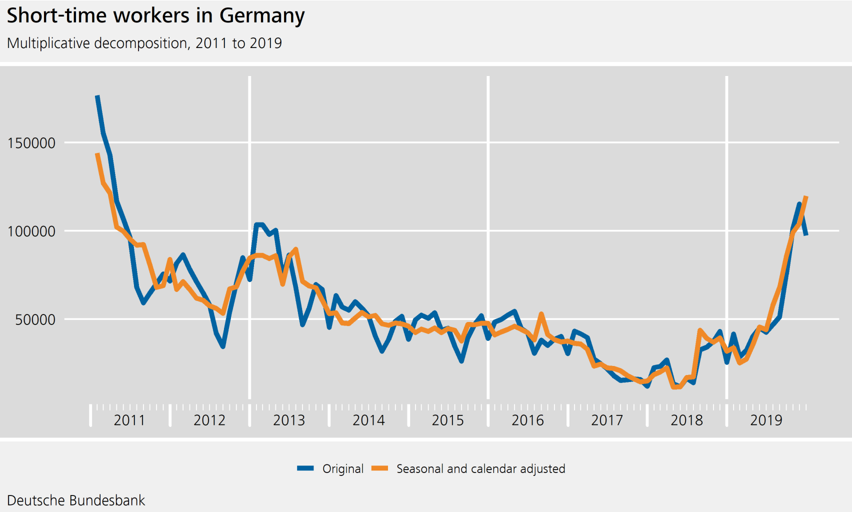
- ESS guidelines on seasonal adjustment, chapter 2.8
- `Methodological note Guidance on treatment of Covid-19 pandemic effect on data`

Covid-19 pandemic: short-time workers

Before the pandemic



Covid-19 pandemic: short-time workers
Standard seasonal adjustment

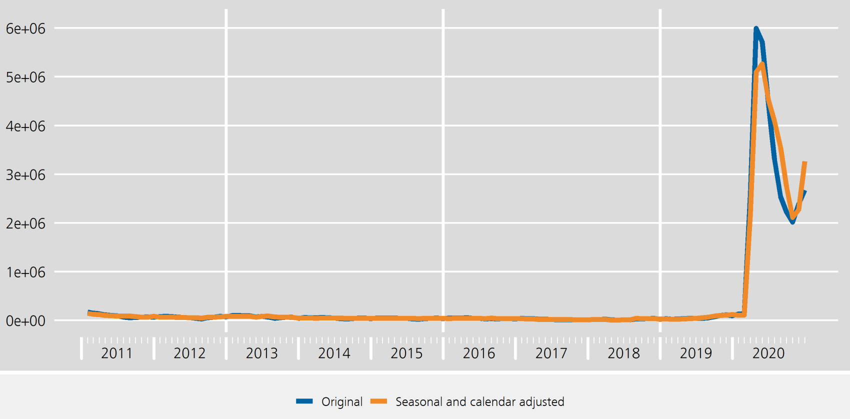


Covid-19 pandemic: short-time workers

Multiplicative assumption does not hold in pandemic

Short-time workers in Germany

Multiplicative decomposition, 2011 to 2020



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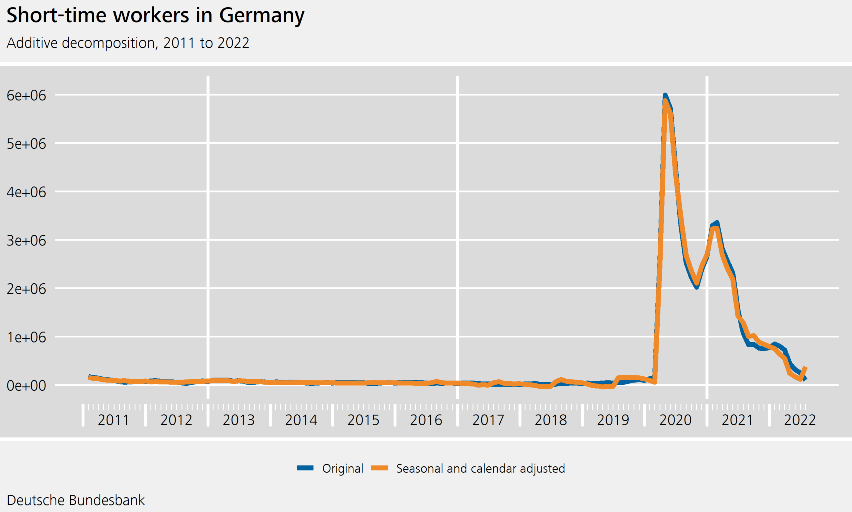
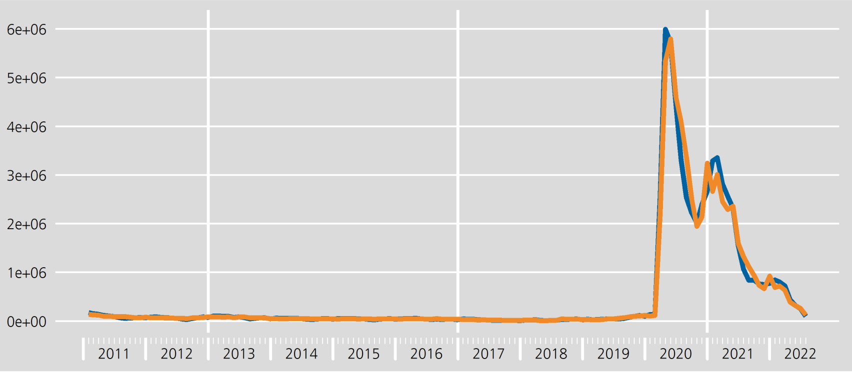
Covid-19 pandemic: short-time workers
Not a solution: negative values in seasonally adjusted series!



Covid-19 pandemic: short-time workers
Non-ideal solution

Short-time workers in Germany

Multiplicative decomposition, 2011 to 2022. Using old seasonal factors for 2020 to 2022



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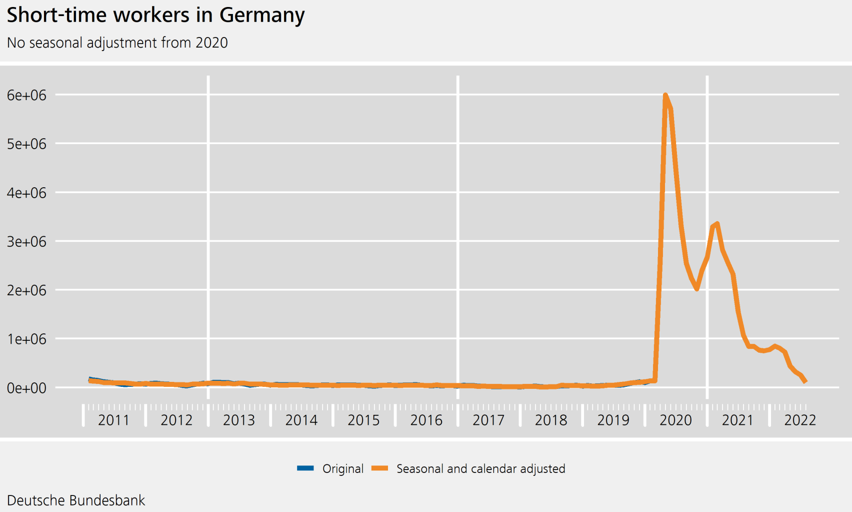
Covid-19 pandemic: short-time workers
Better solution



Exercise

Group work: Do an outlier adjustment for your group's time series

- How do you go about modelling the outliers in this series?
- Does it make a difference whether you use AO or LS to model the series?

Summary

Outlier adjustment is important because

... it improves ARIMA estimation

... it add experts knowledge to model

Covid-19 pandemic

... dominates time series

... makes estimation of seasonal and calendar effectsdifficult

References

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